

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2025-P2-Model)** **Date: 25-03-2026**

Time: 02.30Pm to 05.30Pm **GTA-20** **Max. Marks: 180**

25-03-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2025-P2)_GTA-20_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Rankers Academy JEE

Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2025-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 8)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – IV(Q.N : 9 – 16)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 17 – 20)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 21 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – IV(Q.N : 25 – 32)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 33 – 36)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 37 – 40)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – IV(Q.N : 41 – 48)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

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Negative Marks : -1 In all other cases

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1. Let $\lim\limits_{n\to\infty}\prod\limits_{k=3}^{n}\left(1-\tan^4\frac{\pi}{2^k}\right)=\frac{\pi^3}{m}$, then 'm' is equal to

A) 8 **B)** 16 **C)** 32 **D)** 36

2. If $\int\limits_0^1 \frac{x^4\left(1+x^{10125}\right)}{\left(1+x^5\right)^{2027}}dx=\frac{1}{p}$, then which of the following is **CORRECT** ?

A) Number of ways in which p can be expressed as a product of two relatively prime positive integral factors is 8.

B) Number of ways in which p can be expressed as a product of two relatively prime positive integral factors is 4.

C) Number of ways in which p can be expressed as a product of two positive integral factors is 8.

D) Number of ways in which p can be expressed as a product of two positive integral factors is 16.

3. Which of the following is **INCORRECT**?

A) The minimum value of the function $f(x)=x\sqrt{x}+\frac{1}{x\sqrt{x}}-4\left(x+\frac{1}{x}\right)$ for all permissible real x, is –10.

B) Let f(0) = 0 and $\int\limits_0^2 f'(2t)e^{f(2t)}dt=5$, then the value of f(4) equals ln(11).

C) If $x_1,x_1,x_2,x_3,.....x_{n-1}$ be n zero's of the polynomial $P(x)=x^n+\alpha x+\beta$, where $x_i\neq x_j\forall i,j$ then $(x_1-x_2)(x_1-x_3)(x_1-x_4).....(x_1-x_{n-1})$, equals $n(n-1)x_1^{n-2}$.

D) Let f(x) be a differentiable function such that $f(x)+f'(x)\le 1\forall x\in R$ and f(0) = 0, then the greatest value of f(1) is $1-\frac{1}{e}$.

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4. The number of ordered triplets of sets (A, B, C) such that:

a) $A, B, C \subseteq \{1, 2, 3, ..., 8\}$.

b) $|A \cap B| = |B \cap C| = |C \cap A| = 2$.

c) $|A| = |B| = |C| = 4$.

(Here, $|S|$ denotes the number of elements in the set S) is

A) 45300 **B)** 45360 **C)** 42840 **D)** 5040

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks;
choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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5. $(1+x)^n = C_0 + C_1 x + C_2 x^2 + ... + C_n x^n \left(C_r = {}^nC_r\right)$ and let $\sum_{r=0}^{n} \left(r^2 - 3r + 1\right) C_r = 2^{n-2} f(n)$. If the roots of f(x) = 0 are α and β, then which of the following is/are **CORRECT** ?

A) $3\alpha + 4\beta = 16$ *or* 19

B) $\alpha^2\beta + \beta^2\alpha = 20$

C) For $f(x+1) = 0$, the roots are 1 and 3

D) $(\alpha + 1)$ and $(\beta + 1)$ are roots of the equation $x^2 - 7x + 10 = 0$.

6. Let N be the number of consecutive 0's at the right end of the decimal representation of the product $(1! \cdot 2! \cdot 3! \cdot 4! .....99! \cdot 100!)$. If the remainder obtained when N is divided by 1000 is R, then which is/are **CORRECT**?

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A) Number of positive integral divisors of R is 6.
B) Number of positive integral divisors of R is 8.
C) R is a two digit number.
D) R is a three digit number.

7. Let x, y, z be real numbers with $x \geq y \geq z \geq \frac{\pi}{8}$ such that $x + y + z = \frac{\pi}{2}$ and let $P = \cos x . \sin y . \cos z$, then which of the following is/are **CORRECT** ?

A) Minimum value of P is $\frac{1}{8}$ **B)** Minimum value of P is $\frac{1}{4}$

C) Maximum value of P is $\frac{2+\sqrt{3}}{4}$ **D)** Maximum value of P is $\frac{\sqrt{2}+1}{4\sqrt{2}}$

8. $f'(x):[0,1] \to [p(a), p(b)]$.Given that p is a differentiable function on [a, b] and p(g(x)) = x, a = g(0) and b = g(1). Which of the following is/are **TRUE ?**

A) f(0) + 2 < f(1) **B)** $f(1) \leq 1 + f(0)$

C) $\frac{\int_0^1 f'(x)\,dx}{\int_0^1 g'(x)\,dx} \leq p'(c)$ for some $c \in (a,b)$ **D)** $f(0) + 2 \geq f(1)$

SECTION-III
(NUMERICAL VALUE)

- This section contains **EIGHT (08)** questions stems.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

9. Let A = $\{ n \in N : 'n' \text{ is a 3-digit number}\}$

$B = \{9k + 2 : k \in N\}$

$C = \{9k + \ell; k \in N\}$ for some $\ell (0 < \ell < 9)$

If the sum of elements of the set $A \cap (B \cup C)$ is 61×1800; then 'ℓ' is equal to _____

10. If $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$, then the value of $\det(A^4) + \det\left(A^{10} - (adj(2A))^{10}\right)$ is equal to _____

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11. If $A(Z_1), B(Z_2), C(Z_3)$ are the vertices of ΔABC such that $|Z_1 - Z_2| = 10, |Z_2 - Z_3| = 17$, $|Z_3 - Z_1| = 21$.point $P(Z_4)$ satisfies $|2Z - Z_1 - Z_2| = 10$, then the greatest possible absolute value of $\begin{vmatrix} Z_1 & \overline{Z_1} & 1 \\ Z_3 & \overline{Z_3} & 1 \\ Z_4 & \overline{Z_4} & 1 \end{vmatrix}$ equals

12. The number of triples (x,a,b) where x is a real number and a, b belong to the set $\{1,2,3,4,5,6,7\}$ suchthat $x^2 - a\{x\} + b = 0$, (where { } denotes Fractional Part Function) are _____

13. For the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (where a > b > 0), if the locus of centroid of the triangle formed by the centre and the points of intersection of chord with ellipse which subtends right angle at the origin is $4a^2b^2\left(x^2b^4 + y^2a^4\right) = K\left(a^2 + b^2\right)\left(x^2b^2 + y^2a^2\right)^2$, the value of 4K is equal to

14. A_1 and A_2 are the vertices of the conic $C_1 : 4(x-3)^2 + 9(y-2)^2 - 36 = 0$ and point P is moving in the plane such that $|PA_1 - PA_2| = 3\sqrt{2}$, then the locus of P is another conic C_2. If D_1 denotes the distance between foci of conic C_2. D_2 denotes the product of perpendicular distances from the points A_1, A_2 upon any tangent drawn to the conic C_2 and D_3 denotes length of tangent drawn from any point on auxiliary circle of conic C_1 to the auxiliary circle of the conic C_2, then $\left(\frac{D_1D_2}{D_3^2}\right)^2$ is equal to _____

15. Let $A(\bar{a}), B(\bar{b})$ and $C(\bar{c})$ are the vertices of an acute angled triangle ABC. $D(\bar{d}), E(\bar{e})$ and $F(\bar{f})$ are lying on $\overline{BC}, \overline{CA}$ and $\overline{AB}$ respectively such that $\overline{AD}$ is perpendicular to $\overline{BC}$, $\overline{BE}$ is perpendicular to $\overline{CA}$ and $\overline{CF}$ is perpendicular to $\overline{AB}$ and $P(\bar{p}), Q(\bar{q})$ and $R(\bar{r})$ are the point of intersections of the angular bisectors of angles A, B, C with the sides BC, CA and AB respectively. If the centroid of triangle ABC lies on $\overline{PQ}$, then the value of $\frac{|\overline{AD}| + |\overline{BE}|}{|\overline{CF}|}$ is equal to _____

16. If $x^2 - x - 1$ divides $ax^{12} + bx^{11} + 1 = 0$ where $a, b \in Z$ then the value of $|b + a|$ is _____

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PHYSICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

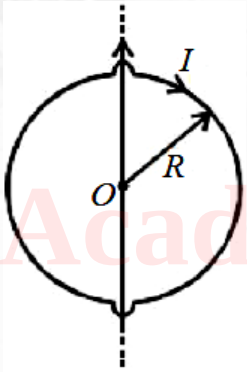
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17. A student is calculating the thickness of a single sheet of paper. She measures the thickness of a stack of 80 sheets with vernier calipers, and finds the thickness to be 1.27 cm. To calculate the thickness of a single sheet she divides the total thickness of sheets with number of sheets. Which of the following answer has the correct number of significant digits?

A) 0.15875 mm **B)** 0.159 mm **C)** 0.16 mm **D)** 0.2 mm

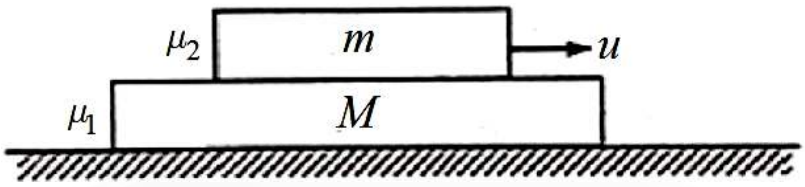
18. A circular conducting loop of radius R carries a current I. Another straight infinite conductor carrying current I passes through the diameter of this loop as shown in the figure. The magnitude of force exerted by the straight conductor on the loop is



A) $\pi\mu_0 I^2$ **B)** $\mu_0 I^2$ **C)** $\frac{\mu_0 I^2}{2\pi}$ **D)** $\frac{\mu_0 I^2}{\pi}$

19. Two long blocks m = 1 kg and M = 3 kg are placed as shown. The coefficient of friction between the blocks is $\mu_2 = 0.8$ and between the floor and M is $\mu_1 = 0.1$. Initially M is at rest and m is given a speed $u = 28 m/s$. Mechanical energy dissipated out of the system of (m + M) due to friction between the blocks is ____ times the loss due to friction at ground. $\left(g = 10 m/s^2\right)$

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A) 4 **B)** 5 **C)** 6 **D)** 7

20. An air bubble with a diameter of a few centimeters was injected into the bottom of a giant closed vertical cylinder that had a height of 10 m and was, (apart from the bubble) completely full of water. Just after injection, the water pressure at the bottom of the cylinder was 1.1 atm. The air bubble rose, and a little bit later it reached the top of the closed container. Assume constant temperature process. Find the pressure (nearly) of the water at the bottom of the cylinder after the air bubble had risen.

For the sake of definiteness, assume that the hydrostatic pressure of 10 m of water is exactly 1 atm and water is highly incompressible.

A) 1 atm **B)** 1.1 atm **C)** 2.1 atm **D)** 2 atm

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

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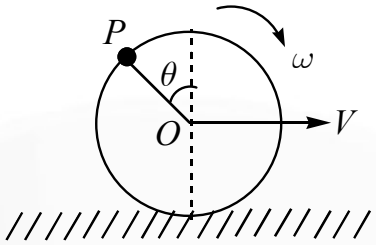
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choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

21. A disc of radius R rolls on a horizontal surface with linear velocity V and angular velocity ω. There is a point P on circumference of disc at angle θ with upward vertical diameter measured anticlockwise (see figure), which has a vertical velocity. Here θ is equal to

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A) $\pi+\sin^{-1}\frac{V}{R\omega}$ **B)** $\frac{\pi}{2}-\sin^{-1}\frac{V}{R\omega}$ **C)** $\pi-\cos^{-1}\frac{V}{R\omega}$ **D)** $\pi+\cos^{-1}\frac{V}{R\omega}$

22. A human body has a surface area of approximately $1m^2$. The normal body temperature is 10K above the surrounding room temperature T_0. Take the room temperature to be $T_0=300K$, for $T_0=300K$, the value of $\sigma T_0^4=460Wm^{-2}$. (assume the body to be black body.)

A) The net rate of energy loss by the body in 1 second is close to 50 joule for the given difference of temperature.

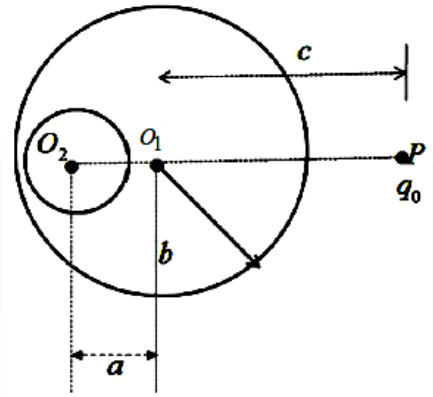
B) If the surrounding temperature reduces by a small amount $\Delta T_0 << T_0$, then to maintain the same body temperature the same (living) human being need to radiate $2\sigma T_0^3\Delta T_0$ more energy per unit time (through some chemical reactions in the body)

C) Reducing the exposed surface area of the body (e.g by curling up) allows humans to maintain the same body temperature while reducing the energy lost by radiation.

D) If the body temperature rises significantly then the peak in the spectrum of electromagnetic radiation emitted by the body would shift to shorter wavelengths.

23. A conducting sphere of radius 'b' has a spherical cavity with its centre displaced by 'a' from centre of sphere. A point charge q is placed at the centre of cavity (O_2). 'Q' charge is given to conducting sphere and charge q_0 is placed at a distance 'c' from the centre (O_1) of sphere such that O_1, O_2 and P are collinear. (assume potential at infinity to be zero)

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THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1
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NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
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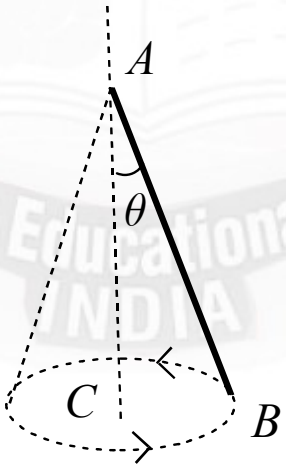
A) Charge distribution on inner surface of cavity is uniform.

B) Potential of conductor is $\left(\frac{q_0}{4\pi\varepsilon_0 c}+\frac{Q+q}{4\pi\varepsilon_0 b}\right)$

C) Charge distribution of outer surface of conducting sphere is non–uniform.

D) Intensity of electric field at O_1 is $\frac{1}{4\pi\varepsilon_0}\frac{q_0}{c^2}$

24. A uniform rod of length 4 m and mass $2\sqrt{2}\,kg$ revolves with constant angular velocity ω about a vertical axis through a smooth joint at fixed point A at one extreme end of the rod so that it describes a cone of semi vertex angle $\theta = 53^\circ$ as shown in the figure, Choose the correct statement (s). $\left(g = 10\,m/s^2\right)$



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A) The value of angular velocity is 2.5 rad/s

B) The angle between hinge reaction at point A and vertical axis AC is 45°

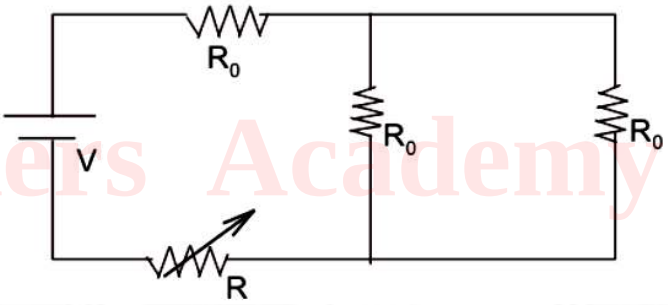
C) The magnitude of hinge reaction at point A is 40N

D) The angular momentum of the rod is constant

SECTION-III
(NUMERICAL VALUE)

- This section contains **EIGHT (08)** questions stems.
- The answer to each question is a **NUMERICAL VALUE.**
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- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
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- **Zero Marks:0** in all other cases

25. In the circuit shown, the value of R for which power dissipated in it will be maximum is $\left(\frac{aR_0}{b}\right)$ then $(a+b)$is? (take a, b values in between the integers 1 and 9) (consider the battery is ideal)



26. Suppose a hypothetical magnetic field $\vec{B} = B_0\hat{\mu}_r$ exists in space above a point on the earth surface (origin) where $\hat{\mu}_r$ is a unit vector directed radial outward from origin. Origin is on surface of earth. A light charged particle has to perform uniform circular motion in the combined uniform (vertical) gravitational field of earth and magnetic field with speed v and radius r. Height of the plane of motion from earth surface will be $h = \frac{nv^2}{g}$. Find n

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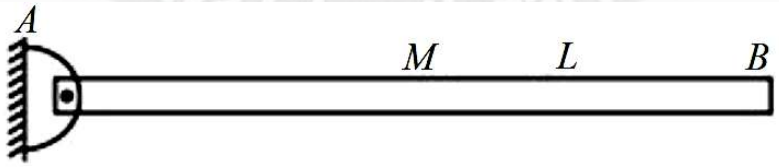
27. Two sound waves $y_1 = A\sin(200\pi t)m$ and $y_2 = A\sin(204\pi t)m$ are interfering at a point where stationary detector is placed which can detect waves of amplitude greater than or equal to A. In a time interval of 12 seconds, find the total duration (in sec) in which detector is active?

28. A vessel is quarter filled with a liquid of refractive index μ. The remaining parts of the vessel is filled with an immiscible liquid of refractive index $3\mu/2$. The apparent depth of the vessel is 50% of the actual depth. The value of μ is $\frac{n}{n-1}$. Find the value of n?

29. A point–like radioactive sample of decay constant λ initially at a distance r_0 from a radiation counter starts moving towards the counter. If the count number rate recorded by the counter remains constant with respect to time, then the speed (in m/s) of the radioactive sample as function of time is given by $v = x_0 e^{-\frac{\lambda t}{2}}$ find the value of x_0 if $r_0 = 2m$, $\lambda = 1\ \text{sec}^{-1}$.

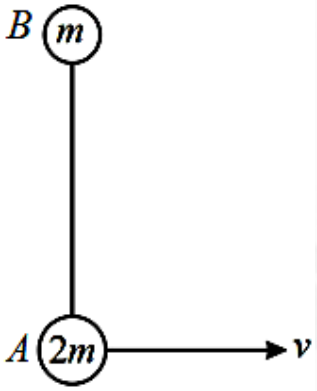
(Ignore the dimensions of the counter, assume it as a point)

30. A thin uniform rod of mass M and length L is free to rotate in vertical plane about a horizontal axis passing through one of its ends. The rod is released from horizontal position shown in the figure. The shear stress developed at the centre of the rod immediately after it is released is $\frac{Mg}{(4A \times n)}$. Then n is _____ [Take cross sectional area of the rod is A, for calculation of moment of inertia you can treat it to be very thin]

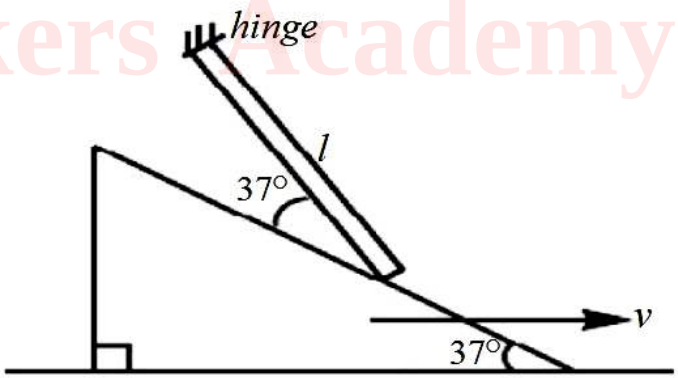


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31. Two small masses A and B connected with an inextensible string of length $l = 1m$ lie on a smooth horizontal plane. A is given a velocity of $v = 3m/s$ along the ground perpendicular to line AB as shown in fig. The tension (in N) in string during their subsequent motion is (take $m = 5kg$)



32. In the figure shown, a wedge is being moved on a horizontal surface with a velocity 'v' as shown. At the instant shown, if angular velocity of the rod of length l (about hinge) touching the wedge as shown is $\frac{xv}{4l}$, the value of x? $\left(take \sin 37^\circ = \frac{3}{5}\right)$



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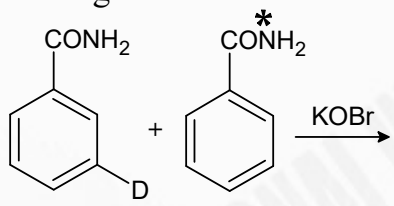
CHEMISTRY **Max Marks: 60**

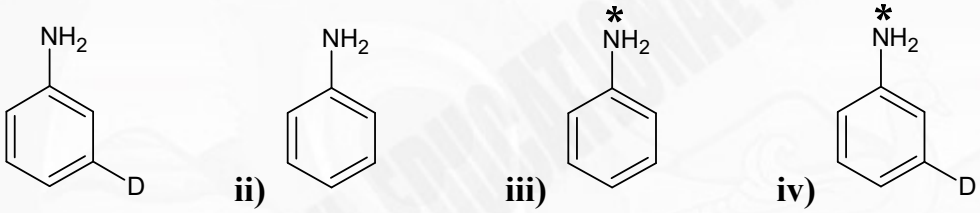
SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases

33. Predict the product in the following reaction





i) **ii)** **iii)** **iv)**

A) i and ii only **B)** i and iii only **C)** ii and iii only **D)** i and iv only

34. A solid consists of atoms X arranged in a cubic close-packed (ccp) lattice. All the octahedral interstitial sites are completely occupied by atoms Y, and all the tetrahedral interstitial sites are completely occupied by atoms Z.Assume that the presence of Y and Z atoms does not cause any distortion in the original ccp arrangement of X atoms.What fraction of the body diagonal of the unit cell remains unoccupied by any atom?

A) 0.26 **B)** 0.24 **C)** 0.76 **D)** 0.81

35. Consider the following statements related to Group 16 elements and their oxidation states in stable binary compounds under normal conditions:

i) No element of Group 16 other than oxygen can exhibit oxidation states of +1 and −1.

ii) S, Se and Te shows +6 oxidation state with fluorine.

iii) On moving down the group, the relative stability of the +4 oxidation state compared to the +6 oxidation state increases.

iv) Oxygen exhibits the −2 oxidation state in all of its binary compounds with halogens, excluding peroxo species.

Which of the following options is correct?

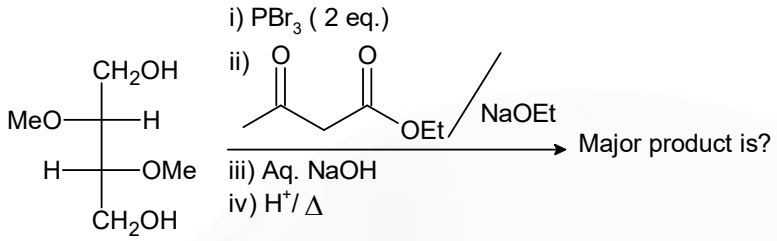
A) All statements (i)–(iv) are correct **B)** Only statement (ii) is correct

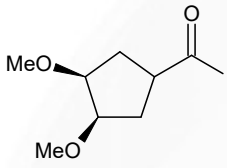
C) Statements (ii) and (iii) are correct **D)** Statements (i), (ii), and (iv) are correct

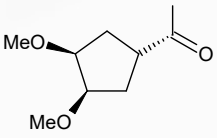
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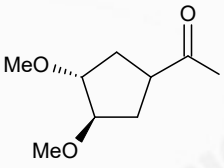

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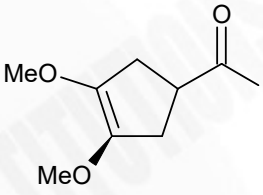
36.



A) 

B) 

C) 

D) 

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

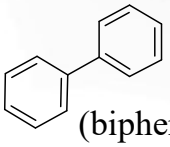
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

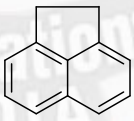
Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

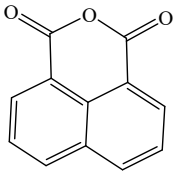
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks;
choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

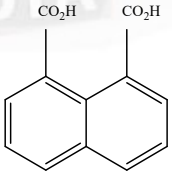
37. Compounds **X** and **Y** are hydrocarbons having the same composition. They contain 93.5% of carbon, and the molecular mass do not exceed 200 $g.mol^{-1}$. **X** and **Y** do not decolourise bromine water, however, react with bromine in the presence of iron. Moreover, only **Y** can be oxidized when heated with $KMnO_4$ solutions, to form B with molecular mass of 216 $g.mol^{-1}$.

Select the appropriate representations

A)  (biphenyl) = X

B)  = Y

C)  = B

D)  = B

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38. A mixture of two different substances A and B undergoes simultaneous first-order reactions to form the same product C. The rate constants are $k_1 = 2\times10^{-2}\,\text{min}^{-1}$ for $A \rightarrow C$ and $k_2 = 5\times10^{-3}\,\text{min}^{-1}$ for $B \rightarrow C$. After 30 minutes from the beginning, the mole percentage of C is found to be 25%. Assume that initially there was no C.

Given: $e^{0.6} = 1.82$, $e^{0.15} = 1.16$, $e^{1} = 2.71$, $e^{0.25} = 1.25$.

Which of the following statements is/are correct?

A) Initially, the mole percentage of A is nearly 36.7%.

B) Initially, the mole percentage of B is nearly 36%.

C) Mole percentage of C after 50 minutes is nearly 35.4%.

D) Mole percentage of C after 50 minutes is nearly 42.54%.

39. For each of the following species of the type XZ_4 (X is the central atom), select all the statements that are correct. A species is considered square if the four Z atoms lie at the corners of a square around X (ignore the positions of lone pairs).

Species: $XeF_4, SiF_4, BF_4^-, BrF_4^-, \left[Cu(NH_3)_4\right]^{2+}, [FeCl_4]^{2-}, [PtCl_4]^{2-}$

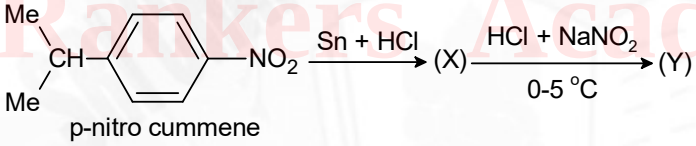
A) Exactly four species have a square planar arrangement of Z atoms around X.

B) Exactly three species are both square planar and diamagnetic.

C) Exactly one species is tetrahedral and paramagnetic.

D) Exactly two species are square planar and paramagnetic.

40.



p-nitro cummene

Y is treated with H_3PO_2 to form R

$$(R) \xrightarrow[(ii)NaOH+I_2]{(i)CH_3COCl/AlCl_3} (A)+(B) \quad \downarrow \text{Yellow ppt}$$

$$(A) \xrightarrow[(iii)(R)/AlCl_3]{\substack{(i)H_2O/H^+ \\ (ii)SOCl_2}} \text{Product}(L)$$

The correct statement (s) about the product (L) is

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A) Product (L) on reaction with Tollen's reagent gives silver mirror but no red precipitate with Fehling's solution.

B) Product (L) on reaction with $H_2N-NH_2 / OH^-, \Delta$ gives diphenyl methane.

C) Product (L) on reaction with HCN gives a compound which contains a chiral centre.

D) Product (L) does not give positive Fehling's test.

SECTION-III
(NUMERICAL VALUE)

- This section contains **EIGHT (08)** questions stems.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks:* +4** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

41. An electrochemical cell operates in an alkaline medium where aluminium metal (Group 13) acts as a sacrificial anode and silicon (Group 14) is deposited at the cathode from an aqueous silicate solution.

Half-reactions:

Anode: $A\ell \rightarrow A\ell^{3+} + 3e^-$

Cathode: $SiO_3^{2-} + 4H_2O + 4e^- \rightarrow Si + 8OH^-$

During electrolysis: Mass of Al anode decreases by 0.270 g.

Only 80% of electrons released at the anode participate in the cathodic reaction shown above.

Total solution volume remains 1.0 L throughout.

Initially, the solution contains 0.020 mol of SiO_3^{2-} and no excess strong base.

Assume all generated OH^- remains in solution and ignore hydrolysis of aluminium ions.

Take M(Al) = 27.

Given: log 48 = 1.68.

Calculate the final pH of the solution.

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42. A sample of platinum metal of mass 1.00 g is present in the form of non-porous spherical particles, each having radius $r = 1.0 \times 10^{-6} m$. The density of platinum metal is $20\, g\, cm^{-3}$. Assume that the volume calculated using density corresponds to the actual volume of platinum metal only, and void spaces between particles are ignored. Hydrogen gas is passed over platinum at 300 K and undergoes dissociative chemisorption to form a complete monolayer: $H_2(g) \rightarrow 2H(ads)$.

Each adsorbed hydrogen atom occupies a surface area of $5.0 \times 10^{-20} m^2$.
All the adsorbed hydrogen is then completely removed and used in an ideal electrochemical cell where: $H_2 \rightarrow 2H^+ + 2e^-$
Given:

Avogadro number, $N_a = 6.0 \times 10^{23} mol^{-1}$

Faraday constant, $F = 1.0 \times 10^5 C mol^{-1}$

Calculate the total charge (in coulombs) that can be obtained.

43. For the equilibrium reaction: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$.

An equilibrium mixture is prepared by starting with 1 volume of N_2 and 3 volumes of H_2. At a total pressure of 0.5 atm, the fraction of ammonia by volume at equilibrium is 0.20. Assuming the temperature remains constant, at what total pressure (in atm) will the fraction. of ammonia at equilibrium become 0.70?

44. A frictionless piston–cylinder at T = 300 K contains a well-mixed ideal liquid solution of two volatile liquids A and B. At 300 K, $P_A^0 = P_B^0 = 1.0\, atm$. Hence, if any liquid remains, the total vapour pressure above the solution is constant at P = 1.0 atm. Initially, the total volume is $V_1 = 2.4 L$ and the cylinder contains 0.20 mol of the liquid solution (liquid volume negligible). The system is expanded reversibly and isothermally until $V_2 = 14.4 L$.

Use: $R = 0.08\, L.atm.mol^{-1}.K^{-1}, 1 L.atm = 0.1 kJ, l\,n\,2 = 0.7$

Total work done by the system (in kJ) is _____

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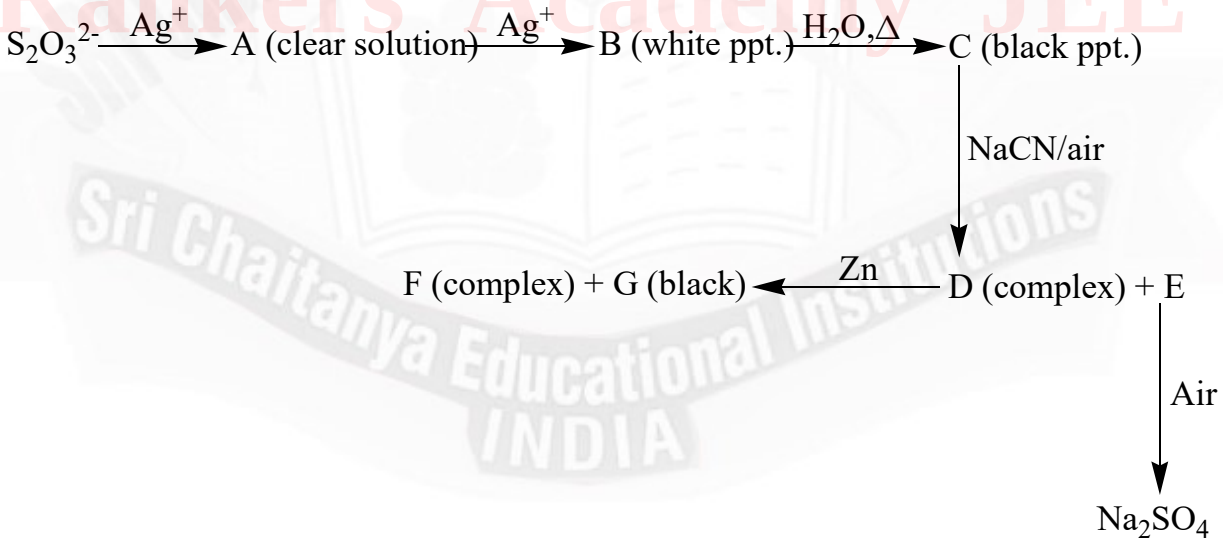
45. An alcohol (A) on heating with concentrated H_2SO_4 gives an alkene (B). Compound (B) reacts with $Br_2 / CC\ell_4$ and the resulting vicinal dibromide undergoes double dehydrohalogenation with $NaNH_2$ (excess) followed by acid workup to form compound (C). Compound (C) on hydration with dilute H_2SO_4 in the presence of $HgSO_4$ gives compound (D). Compound (D) can also be obtained by oxidation of A using PCC, or by dry distillation of calcium acetate. If the molecular mass of C is $(Y \times 10)$, find Y.

46. Gas 'A' reacts with $C\ell_2(g)$ in the presence of charcoal to form B. Elemental phosphorus (P_4) reacts with B to regenerate A(g) and produce C. Compound C reacts with CH_3COOH to give an organic compound D (along with other products). A sample of 7.85 g of D reacts with excess NH_3 in the presence of pyridine to give an organic compound E. Let the mass of E formed be x g. Also, 0.10 mol of B on complete hydrolysis gives acids F and G. Let the mass of $NaOH$ required to neutralize the mixture of F and G be y g.

Find the value of (x + y).

(Atomic masses: Na = 23, Cl = 35.5, S = 32, P = 31, O = 16, H = 1, C = 12, N = 14)

47. In the following reaction sequence in aqueous solution:



Answer the following:

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Answer the following:

i) Number of thiosulphate ions in A is x.

ii) G dissolves in KCN in the presence of air to form a complex compound **L**. y isomers exist for **L**.

iii) E reacts with sodium nitroprusside to form a complex of the type $Na_{\gamma}\left[Fe(CN)_{\delta}NOS\right]$.

Find the value of $(\delta+\gamma+x)/y$.

48. For manganese oxyhydroxide MnO(OH), the following thermochemical data (per 1 mol of MnO(OH)) are known:

1) $MnO(OH)+H_2 \rightarrow MnO+H_2O(g)$ $\Delta H_1=-31.4kJ$

2) $MnO(OH)+\frac{1}{2}O_2 \rightarrow MnO_2+\frac{1}{2}H_2O(g)$ $\Delta H_2=-41.3kJ$

3) $2MnO(OH) \rightarrow Mn_2O_3+H_2O(g)$ $\Delta H_3=-1.0kJ$

4) $3MnO(OH) \rightarrow Mn_3O_4+\frac{3}{2}H_2O(g)$ $\Delta H_4=+16.2kJ$

Also given: $H_2+\frac{1}{2}O_2 \rightarrow H_2O(g), \Delta H=-241.8kJ\,mol^{-1}$

Using the above data, determine the absolute enthalpy change (in kJ) for transformation between Mn_2O_3 and Mn_3O_4 that involves only one mole of oxygen gas is _____

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